

How Refrigerants Are Recycled

As the EPA continues to act upon its ban on gases such as R-22, refrigerant supply goes down but the demand for it stays the same. Many corporations, households and automobiles continue to use the same heating and cooling equipment that need refrigerants to function. Because of this, the prices for safe or “greener” refrigerants go up. This is an important thing to note especially if your company is enforcing stricter budgets.

If you are looking for the most cost-effective way to stay compliant with EPA regulations, you should consider getting your old and harmful refrigerants recycled. But before you do, take a moment to understand how refrigerants are recycled so you know the value of the services you are paying for.

Recycling vs. Reclamation vs. Recovery: What's the difference?

There are different ways to deal with old refrigerants, namely: refrigerant recycling, reclamation or recovery.

Recovering refrigerants – Recovery is done when the HVAC equipment using the refrigerant needs to be tested or repaired. Recovery is the process of taking out the refrigerant from the system using it and then transferring it to a temporary and safe container. The refrigerant used is not changed in any way, so if your refrigerant is non-compliant, this is not the service for you.

Recycling refrigerants – Recycling refrigerants is done by first removing the refrigerant from the system containing it, and then sending it through equipment that will “clean” it. Cleaning refrigerants entails removing the moisture and particles that make it toxic. Once recycling is done, the refrigerant can be put back in its original container, or it can be stored in a different container for future use in a different system.

Reclaiming refrigerants – Reclaiming refrigerants goes a step beyond simply recycling them by making sure that all impurities in the refrigerant are removed, thus returning it into its most pure state. This is according to standards set in the ARI-700, or the standards used for never been used refrigerants. Refrigerant reclamation makes use of more technical equipment, which also makes its services more expensive than recycling refrigerants.

How are refrigerants recycled?

Now that you know the difference between recycling and reclaiming refrigerants, you can better understand what goes on when the refrigerant is sent through the equipment that reduces the contaminants found in it.

First, the refrigerant is sent to a compressor that sucks the refrigerant out of its original container. Then it gets sent to various types of equipment that dries and filters it. The driers take out any moisture that could damage the cooling system of your HVAC equipment. The filters take out any minuscule particles that could cause clogs and further damage to your HVAC equipment. After passing through the driers and filters, the newly recycled refrigerant is then stored in a new tank. You can then use the recycled refrigerant or save it for future use.

Who can recycle refrigerants?

The entire refrigerant recycling process is highly technical and prone to many hazards. When refrigerants are mishandled, they can cause serious setbacks, from environmental contamination to disease.

This is why only licensed technicians can handle equipment that recycles refrigerants. The equipment used for refrigerant recycling must be approved and certified by the Air-Conditioning, Heating, and Refrigeration Institute (AHRI). Failure to comply may lead to penalties or further damage to your current equipment.

Section 608 of the Clean Air Act requires all companies that use heating and cooling equipment with refrigerants to perform the necessary evacuation or sanitation of any old or unused refrigerants. Failure to comply with protocol may lead to suspension of the license to operate along with hefty penalties or fines.

If you are looking for a licensed team who can handle certified equipment to recycle your old refrigerants, contact [Refrigerant Services LLC](#). Not only will we ensure that your refrigerants are handled, recovered and recycled properly, we will also arrange the [pick-up and delivery](#) at your location. Call us at 844-PURECFC (787-3232) to know more about our refrigerant services!